SEMICONDUCTOR ELECTRONICS

FORMULA SHEET | JEE MAINS | Let's Play With Physics

1. ENERGY BANDS & CLASSIFICATION

Material	Band gap E_g	Example (E_g in eV)
Conductor	≈ 0 (bands overlap)	Cu, Ag
Semiconductor	< 3 eV	Si 1.1, Ge 0.7
Insulator	> 3 eV	Diamond 5.4

Max wavelength that can create an e–h pair:

$$\lambda_{\max} = \frac{hc}{E_g} \qquad \lambda\,[\text{nm}] = \frac{1240}{E_g\,[\text{eV}]}$$

Si: $\lambda_{\max} \approx 1127$ nm Ge: $\lambda_{\max} \approx 1771$ nm

⋆ Common Pitfall

- At 0 K a pure semiconductor behaves as an **insulator**
- Semiconductor resistivity **decreases** with T (negative temperature coefficient) — opposite to metals
- Photon with $E < E_g$ (i.e. $\lambda > \lambda_{\max}$) cannot create an electron–hole pair, whatever the intensity

2. INTRINSIC & EXTRINSIC SEMICONDUCTORS

Intrinsic (pure): $n_e = n_h = n_i$

Mass action law (thermal equilibrium, pure or doped):

$$n_e\, n_h = n_i^2 \qquad n_i \propto T^{3/2} e^{-E_g/2k_BT}$$

	n-type	p-type
Dopant	Pentavalent (P, As, Sb)	Trivalent (B, Al, Ga, In)
Impurity	Donor (N_D)	Acceptor (N_A)
Majority	$n_e \approx N_D$	$n_h \approx N_A$
Minority	$n_h = n_i^2/N_D$	$n_e = n_i^2/N_A$
Level	Just below CB	Just above VB

⋆ Common Pitfall

- Both n-type and p-type crystals are **electrically neutral**
- Majority carriers come from doping; minority carriers from thermal generation
- More doping ⇒ fewer minority carriers ($n_{\text{minority}} = n_i^2/N$)
- Donor ionisation energy: ~ 0.01 eV (Ge), ~ 0.05 eV (Si)

3. CONDUCTIVITY & MOBILITY

$$\mu = \frac{v_d}{E}\ \ [\text{m}^2\text{V}^{-1}\text{s}^{-1}] \qquad I = I_e + I_h$$

$$\sigma = \frac{1}{\rho} = e\,(n_e\mu_e + n_h\mu_h)$$

Intrinsic: $\sigma_i = e\,n_i(\mu_e + \mu_h)$ n-type: $\sigma \approx e\,N_D\,\mu_e$

⋆ Common Pitfall

- $\mu_e > \mu_h$ always (holes move by bond-to-bond hopping)
- Holes and electrons drift oppositely, but their **currents add**
- Current density: $j = e(n_e v_e + n_h v_h)$

4. p–n JUNCTION

- **Diffusion current** (majority, p→n for holes) vs **drift current** (minority). At equilibrium they are equal and opposite ⇒ net current = 0
- Depletion layer: immobile ions only, width $\sim 10^{-6}$ m
- Barrier potential: $V_0 \approx 0.7$ V (Si), ≈ 0.3 V (Ge)

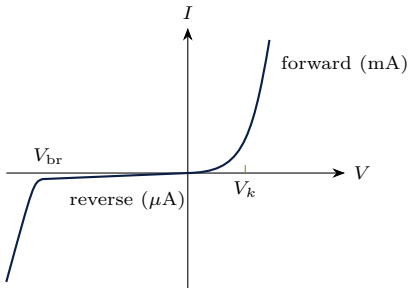
Field in depletion layer: $E \approx \dfrac{V_0}{d}$ (direction n → p)

	Forward	Reverse
Connection	p → (+)	p → (−)
Barrier	$V_0 - V$	$V_0 + V$
Depletion width	decreases	increases
Current	mA (majority)	μA (minority)
Resistance	low	very high

⋆ Common Pitfall

- Diode is **forward biased** if $V_p > V_n$ — check potentials first in every circuit problem
- Field in the depletion layer points from n-side to p-side

5. DIODE I–V CHARACTERISTICS



Dynamic resistance: $r_d = \dfrac{\Delta V}{\Delta I}$

Diode + series R: $I = \dfrac{V - V_k}{R}$ $(V > V_k)$

Ideal diode: forward → short ($R = 0$); reverse → open ($R = \infty$). Knee voltage $V_k \approx 0.7$ V (Si), 0.3 V (Ge).

⋆ Common Pitfall

- Diode is **non-ohmic**; r_d changes along the curve (read slope from the graph)
- Reverse saturation current depends on temperature, not on reverse voltage (until breakdown)

6. RECTIFIERS

	Half-wave	Full-wave
Diodes	1	2 (centre-tap) / 4 (bridge)
$V_{dc} = V_{avg}$	V_m/π	$2V_m/\pi$
I_{rms}	$I_m/2$	$I_m/\sqrt{2}$
Output freq.	f	$2f$
Ripple factor	1.21	0.48
η_{max}	40.6%	81.2%
PIV	V_m	$2V_m$ (CT), V_m (bridge)

For 50 Hz mains: ripple frequency = 50 Hz (HWR), 100 Hz (FWR)

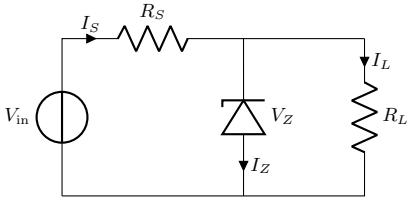
Capacitor filter works well when $RC \gg T$

⋆ Common Pitfall

- In a centre-tap rectifier V_m is the peak of **each half** of the secondary
- Output frequency is doubled only in full-wave rectification
- Larger load R or larger $C \Rightarrow$ smaller ripple

7. ZENER DIODE AS VOLTAGE REGULATOR

Heavily doped, operated in **reverse breakdown**; V_Z stays constant.



$$I_S = \frac{V_{in} - V_Z}{R_S}, \quad I_L = \frac{V_Z}{R_L}, \quad I_Z = I_S - I_L$$

Power in zener: $P_Z = V_Z\, I_Z$

Zener is ON only if $\dfrac{V_{in} R_L}{R_S + R_L} \geq V_Z$

⋆ Common Pitfall

- If the condition fails, zener is OFF (open): $I_L = V_{in}/(R_S + R_L)$
- V_{in} rises $\Rightarrow$ I_S and I_Z rise; V_L & I_L stay fixed
- Zener breakdown: heavy doping, field emission, $V_Z \lesssim 6$ V; Avalanche: light doping, higher voltage

8. OPTOELECTRONIC JUNCTION DEVICES

Device	Bias	Principle	I–V
Photodiode	Reverse	Light → e–h pairs → current	3rd quad.
LED	Forward	Recombination → light	1st quad.
Solar cell	None	Photovoltaic emf	4th quad.

Photodiode: $h\nu > E_g$; photocurrent $\propto$ intensity

LED: $\lambda \approx \dfrac{hc}{E_g}$; visible needs 1.8 eV $\lesssim E_g \lesssim$ 3 eV

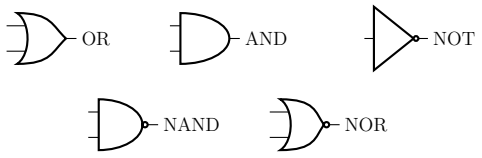
Solar cell: generation, separation, collection; best $E_g \approx 1.5$ eV

Fill factor $FF = \dfrac{P_{max}}{V_{oc}\, I_{sc}}$

⋆ Common Pitfall

- Photodiode is reverse biased because the **fractional** change in minority current due to light is large
- LED reverse breakdown voltage is low ($\sim$ 5 V); Si & Ge are not used for LEDs (indirect gap)
- Solar cell gives emf with no external battery

9. LOGIC GATES



A	B	OR	AND	NAND	NOR	XOR
0	0	0	0	1	1	0
0	1	1	0	1	0	1
1	0	1	0	1	0	1
1	1	1	1	0	0	0

OR: $Y = A + B$ AND: $Y = A \cdot B$ NOT: $Y = \overline{A}$

NAND: $Y = \overline{A \cdot B}$ NOR: $Y = \overline{A + B}$

XOR: $Y = A\overline{B} + \overline{A}B$

De Morgan: $\overline{A + B} = \overline{A} \cdot \overline{B}, \quad \overline{A \cdot B} = \overline{A} + \overline{B}$

Boolean identities: $A + A = A$, $A \cdot A = A$, $A + \overline{A} = 1$, $A \cdot \overline{A} = 0$, $A + 1 = 1$, $A \cdot 0 = 0$, $\overline{\overline{A}} = A$, $A + AB = A$

Universal gates (NAND / NOR)

- NAND/NOR with both inputs joined → NOT
- NAND → NOT = AND; NOR → NOT = OR
- Inverted inputs into NAND = OR; inverted inputs into NOR = AND

⋆ Common Pitfall

- For waveform questions, go interval by interval and write the truth value in each interval
- Simplify the Boolean expression first (De Morgan) before building the truth table

10. JEE SHORTCUTS & MASTER TIPS

1. λ [nm] $= 1240/E_g$ [eV] (photodiode cut-off / LED colour)
2. $n_e n_h = n_i^2$ — use it for every minority-carrier question
3. $\sigma = e(n_e\mu_e + n_h\mu_h)$
4. HWR: $V_{dc} = V_m/\pi$; FWR: $2V_m/\pi$; FWR output freq $= 2f$
5. Zener: first remove the zener and check if $V_{R_L} \geq V_Z$
6. Diode in circuit: replace forward-biased ideal diode by a wire, reverse-biased by a gap

⋆ Master Confusion List

1. **Bias:** Photodiode & Zener → reverse; LED → forward; solar cell → no bias
2. **Neutrality:** n-type is NOT negatively charged
3. **Temperature:** $T \uparrow \Rightarrow$ semiconductor $R \downarrow$, metal $R \uparrow$
4. **Depletion width:** forward $\downarrow$, reverse $\uparrow$
5. **PIV:** centre-tap $2V_m$, bridge V_m, half-wave V_m
6. **NAND/NOR** are universal; AND/OR/NOT are not

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